

Boundary-Uncertainty Active Learning for Efficient Segmentation of Robotic Performance Trajectories

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Abstract: Robotic art installations generate rich, high-frequency motion data during live performance, yet this data is rarely used by AI that could model, analyse or generatively reproduce the behaviours it captures. Performance recordings are unstructured, variable in length and arrive unlabelled, often requiring segmentation through manual segmentation that is prohibitive at scale. We present an active-learning pipeline that transforms raw recordings from Embrace Angels (2025-), an ongoing robotic art performative project in which a pair of Franka Panda arms embrace audience members, into a stage-segmented dataset suitable for future modelling. The pipeline combines an integrated annotation interface, self-supervised pretraining and a boundary-uncertainty acquisition function. We benchmark segmentation methods spanning zero-label baselines, classical template matching, window-based classical active learning with three base machine learning models and three neural sequence-model variants. A Temporal Convolutional Network (TCN) initialised from self-supervised pretraining achieves a boundary mean absolute error of 5.9 ± 0.9 seconds with a 76% reduction in annotation effort relative to fully manual labelling. Overall, active learning combined with self-supervised pretraining is particularly well suited to performance datasets where intra-class variability is high, class boundaries are inherently subjective and the marginal value of each additional labelled example can be extremely informative.

Keywords: Sequence Segmentation, Boundary-Uncertainty Active Learning, Robotic Performance Data, Embodiment, temporal convolutional networks

1. Introduction

The intersection of robotics and live performance has produced a growing body of artistic work in which robots act as collaborators alongside human performers and audiences. These installations generate high-frequency multi-joint motion records captured during one-off interactions that cannot easily be repeated. Such data is artistically and scientifically valuable as it records how a robot’s behaviour unfolded in response to a real human in a real-world context that synthetic data and laboratory studies struggle to reproduce.

The use of robots in live performance and interactive art has grown into a substantial body of work spanning theatre, dance, and installation contexts. Demers [1] characterises the robotic performer as occupying a distinctive position between tool, character and collaborator and [2] explores similarities between contemporary performing robots and earlier traditions of puppetry. Studies of dance with robots, including work with mixed-ability dancers [3], foreground the felt, qualitative dimensions of human–robot movement interaction over task-completion metrics.

A persistent feature of this work is that it generates substantial archival material such as robot motion logs that documents the artistic performance in detail and from different perspectives. It has been argued to treat this material as data in its own right, both for analysis [4, 5] and as a substrate for generative modelling that might extend the artistic practice [6]. However, the practical machinery for turning performance documentation and any resulting archives into structured datasets remains underdeveloped. Most recent gen-

erative motion datasets such as HumanML3D [7] and KIT-ML [8] are built from controlled motion-capture sessions with scripted prompts, not from the unscripted, one-shot recordings characteristic of performance contexts.

One reason why performance documentation rarely becomes AI training data is due to the unstructured nature of the data, recordings vary in length and structure where the meaningful moments of interaction are not separated by clean state changes but by gradual, often subjective transitions. Furthermore, the volume of recordings produced by even a short artistic performance quickly outpaces what can be feasible to annotate by hand.

Embrace Angels [9], the robotic art project we study here, highlights these challenges. It is an intimate encounter where two participants step onto a red carpet between two Franka Panda arms [10], which then perform a shared group embrace experience. Each arm logs its full state where we capture seven joint positions per arm. Crucially, the stage boundaries are interpretive, there is no objective signal that defines exactly when one segment ends and the next begins. A human annotator watches the rendered motion and makes a judgement informed by the movement.

This paper presents a pipeline that addresses this gap for Embrace Angels, but with techniques that generalise. We have accumulated 188 embrace recordings, each consisting of paired CSV files capturing 7-DOF joint positions for both arms at approximately 877 Hz. To train AI models on this data we first needed to segment each recording into its constituent stages which we define as *entering* (the arms beginning to move), *opening* (preparing to receive the participant), *embracing* (the held contact) and *leaving* (release).

This work makes four contributions: 1) An integrated annotation tool that pairs simulator rendered playback of the robot motion with a per-recording timeline and stage marking. 2) A self-supervised pretraining stage that exploits the unlabelled data. 3) A boundary-uncertainty acquisition function for Active Learning (AL) on sequence segmentation. 4) A benchmark comprising of a range of segmentation methods on the Embrace Angels dataset.

2. Background

2.1. Time-series Segmentation

Segmentation of time series data into contiguous segments is a long-standing problem. Change-point detection methods identify time points at which the underlying distribution changes. For example, binary segmentation [11] with a Radial Basis Function (RBF) cost [12] and Pruned Exact Linear Time (PELT) [13] which provides optimal detection under additive penalties. These approaches are useful because they require no labels and make minimal assumptions about segment content but they detect statistical boundaries which often don't well align with semantically meaningful transitions such as those in artworks.

Dynamic time warping [14, 15] provides an alternative option for segmentation by template matching, where a labelled segment boundaries can be propagated to a new sequence via the warping path. [16] provides an exploration in the context of music and motion alignment.

In video, action segmentation has moved toward neural network based models. Temporal Convolutional Networks (TCN) [17, 18] replaces earlier recurrent approaches by stacking dilated convolutions to achieve long effective receptive fields with parallel training. The MS-TCN architecture [19] extended this with multi-stage refinement specifically for action segmentation and Ishikawa et al. [20] introduced an explicit boundary-detection head trained jointly with frame-wise classification, motivated by the observation that over-segmentation is the dominant error mode in this setting.

2.2. Self-supervised Pretraining

Self-supervised pretraining has emerged as an effective way to exploit unlabelled time-series data. A transformer-based framework [21] pretrains on masked reconstruction, predicting masked input values from surrounding context and demonstrates strong transfer to regression and classification. TS2Vec [22] uses contrastive learning at multiple temporal granularities and TF-C [23] contrasts time-domain and frequency-domain views of the same signal. The use of pretrained representations from unlabelled data to improve a future small-label classifier is the same principle invoked in semi-supervised learning more broadly [24].

2.3. Active learning

Active learning [25] aims to solve the problem of selecting which unlabelled instances to manually label in order to maximise model improvement. Uncertainty sampling [26] involves querying the instances on which the current model is most uncertain remains a competitive baseline. Query-by-committee approaches [27] rank instances by disagreement among an ensemble of models, this idea connects naturally to deep ensembles [28] which provide well-calibrated uncertainty estimates for classifiers. Bayesian active learning by disagreement (BALD) [29] query instances with the highest mutual information between predictions and model parameters, Gal et al. [30] extended this to deep networks using Monte Carlo dropout and Beluch et al. [31] showed that ensemble-based uncertainty often matches or exceeds Bayesian approximations for active learning in image classification.

Active learning for sequence labelling raises important questions as instances are sequences of frames. Settles and Craven [32] provide a foundational analysis for named-entity recognition and related NLP tasks, comparing token-level and sequence-level acquisition functions and finding that the appropriate aggregation depends on the structure of the labels. [33] and [34] extend these analyses to deep sequence taggers. In activity-recognition contexts, active learning to reduce annotation burden for wearable-sensors [35] and deep classifiers found consistent improvements from uncertainty-based acquisition over random selection [36].

3. Active Learning Pipeline

This section describes the components of the pipeline. We treat the segmentation problem as a benchmark, comparing zero-label baselines, classical template and probabilistic methods, window-based classifiers and neural networks. We additionally specify an acquisition function that ranks unlabelled recordings by their expected informativeness that targets the temporal location of stage boundaries, rather than the per-frame label uncertainty conventionally used in active learning for sequence tasks.

3.1. Dataset

The Embrace Angels dataset comprises of 188 embrace recordings accumulated across a public performance run at the Cinekid Festival in Amsterdam in 2025. Each recording consists of a paired set of CSV logs, one per arm, capturing 7-DOF joint positions for both Franka Panda arms at approximately 877 Hz. Recordings vary in length and structure, reflecting the unscripted, one-shot nature of each encounter where no two embraces unfold identically and the duration of each stage depends on the participants' behaviour. We define four interpretively-labelled stages as entering (the arms beginning to move), opening (preparing to receive the participant), embracing (the held contact), and leaving (release), whose boundaries are determined by human judgement rather than any objective signal in the data. This combination of high intra-recording variability due to varying robot operators and different performers and subjective boundaries makes the data representative of performance archives more broadly.

3.2. Annotation Tool

A custom annotation tool has been developed that simulates robot motion, rendered from the raw CSV data using MuJoCo [37]. A timeline highlights the regions between markers, giving the annotator visual feedback on the segmentation so far. Annotators set markers for the corresponding stage to the current position and annotations are saved as JSON files alongside each recording, storing both the frame index and the corresponding CSV row index for each stage marker. The annotation tool makes manual labelling efficient on a per-recording basis, but it is still challenging to know which recordings to annotate. An annotator working through the dataset will spend much of their effort on recordings whose structure is already well-captured by recordings already labelled, while genuinely informative outliers wait to be annotated. The role of the active-learning pipeline is to invert this where from a small set of manual annotations it predicts boundaries on the unlabelled remainder and highlights the recordings whose labels would most improve the model for manual annotation.

3.3. Feature Extraction

The full 62-dimensional per-frame feature vector comprises 42 joint-space features (position, velocity, and acceleration for each of the 7 joints on both arms) and 20 task-space features (per-arm end-effector position and velocity (3 dims each), scalar end-effector speed, reach norm and motion energy (1 dim each), the inter-arm end-effector distance and its time derivative (1 dim each)). Each method is paired with the feature set best suited to its capacity. The classical baselines (heuristic, Pruned Exact Linear Time (PELT), Dynamic Time Warping (DTW) and Hidden Semi-Markov Model (HMM)) operate on the 20-dimensional task-space subset, since they rely on low-dimensional signals. The window-based classifiers and neural sequence models use the full 62-dimensional vector as the classifiers consume fixed-length summary statistics (mean, standard deviation, minimum, maximum, range) over overlapping 500- and 1500-frame windows, while the neural models consume the full per-frame sequence directly.

3.4. Self-supervised Pretraining

A 1D temporal convolutional encoder (kernel size 3, dilations 1/2/4/8, latent dimension 64) is trained on a masked-reconstruction objective over the entire unlabelled dataset. Training windows of 500 frames are sampled across all 188 recordings where for each window, 15% of frames are masked and the encoder is trained jointly with a linear reconstruction head to reproduce the masked values under mean-squared error, evaluated only at masked positions. The pretrained encoder serves two roles, first it produces per-window 64-dimensional latent vectors that are concatenated to summary-statistic features for one variant of the window-based ensemble classifier. Second, it initialises the encoder of the Self-Supervised Learning (SSL) neural networks, which are then fine-tuned with a reduced encoder learning rate. Because pretraining uses no labels, this stage is equivalent to an unsupervised feature-engineering step that exploits the dataset.

The use of SSL is motivated by the modest scale of the labelled data. SSL pretraining on the full unlabelled dataset provides the encoder with representations shaped by the global distribution of embrace motions before any label supervision is applied. Because SSL uses no annotations, it does not constitute label leakage as each recording’s stage annotations remain unseen by any model until that recording enters the labelled set.

3.5. Classical Baselines

Heuristic zero-label segmentation applies fixed thresholds to smoothed task-space signals (motion-energy crossings for entering, low-distance windows for embracing, peak-reach for opening). PELT change-point detection [12] runs binary segmentation with an RBF cost on a three-channel signal (distance, motion energy, reach) and requests three breakpoints. DTW template propagation aligns each query against every labelled template using Sakoe–Chiba-banded DTW on z-scored task-space features and transfers boundaries through the warping path, aggregating across the three nearest templates.

3.6. Window-based Ensemble Classifiers

Three classical base classifiers are tested with different acquisition function. Windows are extracted at two scales (500 and 1500 frames) with stride 250 and summarised by per-feature mean, standard deviation, min, max, and range. The base classifiers are HistGradientBoosting (HGB), Support Vector Machine (SVM) with RBF kernel and probability calibration ($C=10$), and Random Forest (200 trees, max depth 10). Each method trains five instances of the same classifier, each on a different random sample of the labelled recordings. At prediction time we average the five models’ per-window stage probabilities, then run a left-to-right Viterbi pass that forces stages to appear in the correct order. The predicted boundary for each stage is the first window where that stage appears. One variant, HGB+SSL+pseudo, adds the frozen SSL encoder’s latent features to the window summaries.

We additionally evaluate HGB+SSL+pseudo, that combines pseudo-labelling with the frozen SSL latent features. During each active-learning round, the 20% of pool recordings with the lowest ensemble boundary-uncertainty are pseudo-labelled with the ensemble’s predicted boundaries and added to the training set, around 25 recordings in round 1, decreasing to 11 by round 8 as the pool shrinks.

3.7. Neural Networks

Two neural networks predict per-frame stage labels and boundary locations. The TCN model uses the same encoder architecture as the SSL stage with a per-frame stage-classification head. We train the TCN with random encoder initialisation and TCN (SSL-init) with encoder initialised from the SSL-pretrained weights and fine-tuned over 100 epochs. The Bidirectional Long Short-Term Memory (BiLSTM) variant replaces the TCN encoder with a two-layer bidirectional LSTM (hidden 64 per direction) under an identical training protocol. The BiLSTM variant is trained from random initialisation as its recurrent architecture is not compatible with the SSL-pretrained weights. All neural variants use a five-member deep ensemble with recording-level bootstrap. At inference, per-frame stage logits are decoded by the same Viterbi procedure used for the window classifiers.

3.8. Boundary-uncertainty Acquisition Function

Given the per-frame predictions of the above models, our boundary-uncertainty acquisition function decides which unlabelled recordings to annotate next. We train an ensemble of K models, each on a different samples of the currently labelled recordings and apply all K to every recording in the unlabelled pool. Where the ensemble agrees on the location of a stage transition, the recording is already well explained by the existing labels and adds little but where it disagrees, the recording is informative and a manual label would most improve the model. Our acquisition function quantifies this disagreement directly, ranking each pool recording by how much the ensemble members differ in where they place its stage boundaries, rather than by per-frame label uncertainty. The recordings with the greatest disagreement are queried first, labelled by the annotator and added to the training set for the next round. We compare this against uniform random sampling (no-AL baseline) and mean per-window predictive entropy (conventional AL signal for sequence labelling).

$$\mathcal{A}_{\text{bnd}}(r) = \sum_{s \in S} \text{std}_{k \in \{1, \dots, K\}} [b_{k,s}(r)] \quad (3.1)$$

Equation 3.1 defines the boundary-uncertainty acquisition function. Here r is a recording in the unlabelled pool and S is the set of stage transitions to be located. Each of the K ensemble members independently decodes the recording via the Viterbi pass, yielding $b_{k,s}(r)$, the time (in seconds) at which member k places transition s in recording r . For a given transition, $\text{std}_k(b_{k,s}(r))$ is the standard deviation of these K predicted times. Summing over the transitions in S yields a single acquisition score $\mathcal{A}_{\text{bnd}}(r)$ per recording, the total boundary-location disagreement across the whole encounter.

A known limitation of disagreement-based acquisition functions is that they cannot detect errors that all ensemble members share e.g. if all models are wrong in the same way such as misplacing a boundary the training data has never disambiguated, the acquisition score will be low even though the prediction is poor. Diversifying the ensemble at training time mitigates this, in our benchmark bootstrap resampling at the recording level provides the diversity, but shared biases across ensemble members remain a challenge.

3.9. Experimental Protocol

To evaluate each method under realistic labelling budgets, we simulate the active-learning loop from a small seed set up to roughly half the data, we stop here because retaining a 50% pool of never-queried recordings allows an unbiased estimate of held-out generalisation. At each of five random seeds we partition the dataset into a 20% held-out test set, a seed set comprising 10% of the remaining (non-test) recordings and a 70% training pool. All methods receive the same splits. At each of eight subsequent active-learning rounds the acquisition function ranks the current pool, the top 10 recordings are queried and they are added to the labelled set and the model is retrained from scratch. We evaluate on the held-out test set at every round, yielding a learning curve of boundary error against labelling budget. The final budget is 95 labels. We report boundary Mean Absolute Error (MAE), median in seconds, hit-rate at 2.0 second tolerance and per-frame balanced accuracy. 2.0s tolerance was chosen because stage durations are typically tens of seconds and human perception of movement boundaries is coarse at sub-2s scale where errors are visually imperceptible in embrace performances.

Table 1. Final performance across at 95 labelled recordings ($\sim 50\%$ of pool). Mean $\pm$ std over 5 random seeds. Lower is better for MAE and median. Higher is better for hit-rate and balanced accuracy.

	MAE (s)	median (s)	hit-2s	bal. acc.
Heuristic	48.03 ± 4.64	10.52 ± 1.65	0.22 ± 0.01	0.66 ± 0.02
PELT	25.64 ± 3.80	7.31 ± 0.95	0.32 ± 0.02	0.67 ± 0.03
DTW	6.87 ± 0.93	2.78 ± 0.29	0.43 ± 0.03	0.85 ± 0.02
SVM + boundary-unc AL	8.58 ± 0.93	3.58 ± 1.62	0.45 ± 0.05	0.61 ± 0.02
RF + boundary-unc AL	8.24 ± 0.68	4.58 ± 1.14	0.45 ± 0.03	0.56 ± 0.03
HGB + boundary-unc AL	7.80 ± 0.47	5.15 ± 0.68	0.37 ± 0.03	0.74 ± 0.02
HGB + SSL + bnd-unc + pseudo	7.59 ± 0.22	4.81 ± 0.45	0.38 ± 0.02	0.74 ± 0.02
BiLSTM + bnd-unc AL	6.41 ± 0.71	2.47 ± 0.57	0.47 ± 0.05	0.84 ± 0.02
TCN + bnd-unc AL	6.24 ± 1.26	1.99 ± 0.24	0.50 ± 0.03	0.86 ± 0.02
TCN (SSL-init) + bnd-unc AL	5.91 ± 0.89	1.74 ± 0.23	0.53 ± 0.03	0.86 ± 0.02

4. Results

4.1. Method Comparison

Table 1 reports final-budget performance for all methods. TCN (SSL-init) with boundary-uncertainty AL achieves the lowest MAE (5.91 ± 0.89 s), lowest median boundary error (1.74 ± 0.23 s) and the highest per-frame balanced accuracy (0.86 ± 0.02). The TCN (without SSL) and BiLSTM variants performed similarly as all three neural methods reach a per-frame balanced accuracy of 0.84–0.86, above the best classical AL pipeline (0.74). The gap between mean MAE and median boundary error indicates that a small number of outlier recordings dominate the mean. The four stages differ substantially in duration, opening is the shortest with a median duration of 20.5s (range 2–123s), while entering is the longest with a median of 129s embracing and leaving fall between these, with medians of 47.8s and 51.0s respectively. Relative to these durations, a median boundary error of 1.74s corresponds to less than 10% of even the shortest stage and is well within the tolerance required for downstream generative modelling, where sub-second boundary precision is not perceptually meaningful.

The unsupervised baselines are clearly separated with PELT (25.6s) substantially outperforming the heuristic method (48.0s) without using any labels but both remain significantly worse than any supervised method. DTW is the strongest non-neural method (6.87 ± 0.93 s on MAE, 0.85 balanced accuracy), competitive with supervised classical AL despite requiring no model training, reflecting the highly stereotyped nature of the embrace motions.

4.2. Boundary-Uncertainty Active learning

Table 2 shows the contribution of acquisition strategy by holding the base model fixed and varying only the AL signal. Boundary-uncertainty AL outperforms every base model, improving over random sampling by 1.09s (SVM), 0.97s Random Forest (RF), 0.79s (HGB), 0.59s (BiLSTM), 0.12s (TCN) and 0.16s (TCN SSL-init).

Table 2. Boundary MAE in seconds at final budget (95 labels), mean $\pm$ std over 5 seeds.

	random AL	entropy AL	bnd-unc AL
SVM	9.67 ± 1.59	9.48 ± 0.87	8.58 ± 0.93
RF	9.21 ± 0.22	9.33 ± 0.61	8.24 ± 0.68
HGB	8.59 ± 1.74	8.06 ± 0.40	7.80 ± 0.47
BiLSTM	7.00 ± 1.41	6.79 ± 0.47	6.41 ± 0.71
TCN	6.36 ± 2.56	6.86 ± 1.07	6.24 ± 1.26
TCN (SSL-init)	6.07 ± 0.55	7.65 ± 1.39	5.91 ± 0.89

To test statistical significance we compute Wilcoxon paired tests across the 45 comparison pairs available for each base as shown in Table 3. The gain over random sampling is statistically significant on five of six base models (SVM, RF, BiLSTM, TCN and TCN (SSL-init)), with only HGB showing no significant

difference. Mean predictive entropy is a markedly weaker signal as it underperforms random sampling on RF and on both TCN variants, where treating every frame as equally informative appears to actively mislead acquisition.

Table 3. Wilcoxon paired test of boundary-uncertainty AL against random AL on MAE, paired across all 45 (seed $\times$ budget) observations. Significance: $p < 0.05$.

	Δ MAE (s)	p -value
SVM	−1.91	0.0021
RF	−0.51	0.0014
HGB	+0.05	0.1903
BiLSTM	−1.18	0.0143
TCN	−1.50	0.0014
TCN (SSL-init)	−1.00	0.0035

4.3. Label Efficiency

Table 4 characterises the label efficiency of each method across varying labelling budgets. The results show that neural network models reach their final accuracy at 45 labelled recordings (24% of the 188 recordings). The MAE of TCN is 6.2s at both 45 and 95 labels, BiLSTM is 6.4s at both budgets and TCN (SSL-init) improves only marginally (from 6.7s to 5.9s). Therefore, doubling the number of labels from 24% to 50% of the dataset results in no statistically distinguishable improvement on three of the four best-performing methods.

The active-learning signal is most visible when using fewer labels, for example when using SVM, boundary-uncertainty acquisition reduces MAE from 16.9s at 15 labels to 10.9s after a single AL round of 10 queries, while random sampling at the same budget achieves 18.9s, an eight-second difference. Overall, this shows that targeted active labelling combined with an neural sequence models can segment the majority of the dataset from around a quarter of the labels, with the value of additional labels diminishing sharply thereafter.

Table 4. Boundary MAE (seconds) at increasing labelling budgets, mean across 5 seeds. Percentages refer to the fraction of the 188-recording pool labelled.

	15 (8%)	25 (13%)	45 (24%)	65 (34%)	95 (50%)
Heuristic	48.0	48.0	48.0	48.0	48.0
PELT	25.6	25.6	25.6	25.6	25.6
DTW	10.6	9.6	9.6	8.8	6.9
HGB + bnd-unc	9.5	8.3	8.5	8.2	7.8
HGB + SSL + bnd-unc + pseudo	7.6	8.1	7.7	7.9	7.6
BiLSTM + bnd-unc	14.2	8.7	6.4	5.8	6.4
TCN (scratch) + bnd-unc	23.0	10.3	6.2	5.6	6.2
TCN (SSL init) + bnd-unc	22.1	13.3	6.7	6.0	5.9

5. Discussion

The combination of self-supervised pretraining, a neural sequence model and boundary-uncertainty active learning reaches near-best segmentation accuracy at around a quarter of the pool labelled, removing $\sim 75\%$ of the manual effort that fully-supervised annotation would have required. The proposed boundary-uncertainty acquisition function is statistically significant on five out of the six classical and neural classifiers and produces its largest gains in the low-label regime where the choice of which recording to label matters most.

Self-supervised pretraining produces small but consistent improvements although at the smallest budgets SSL initialisation offers no advantage and the pretrained model converges more slowly than training from scratch. TCN (SSL-init) improves over the baseline TCN by 0.33s in mean MAE and 0.25s in median and reaches the highest hit-rate-at-2s of any method (0.53 vs 0.50 for the baseline). For the HGB classifier, concatenating frozen SSL latent vectors to the window summary features (combined with pseudo-labelling)

reduces MAE from 7.80s to 7.59s. These improvements are positive but modest, the larger impact comes from the choice of base architecture and the AL signal. We note that this pattern of modest but consistent SSL gains on small dataset is in keeping with prior reports on time-series SSL [21,22], where the marginal benefit of pretraining depends strongly on the size of the unlabelled pool.

The significance of the boundary-uncertainty gain shrinks as base-model capacity grows, from over a second on the classical classifiers to roughly 0.1–0.2s on the neural models. However, the Wilcoxon tests remain significant on the neural models because they pool across the full learning curve. Table 4 shows that the neural models are highly inaccurate when using the smallest labelled data pool (over 22s MAE at 15 labels) and converge to within 0.2s of their final accuracy by 45 labels so the acquisition signal delivers its benefit early but this advantage narrows as the labelled pool increases.

The strong DTW result reflects the highly stereotyped nature of the embrace motions. However, DTW’s cost scales as $O(n^2 \cdot K)$ where K is the template count, so query-time inference grows linearly with the labelled dataset, a limitation that becomes significant as datasets grow. Neural models additionally produce per-frame stage probabilities suitable for downstream generative conditioning, while DTW returns only boundary indices.

Entropy sampling potentially underperforms random on TCN variants as well-trained neural sequence models produce sharp softmax distributions on most frames, so mean per-frame entropy is dominated by a small tail of ambiguous frames whose location varies across recordings and is not systematically informative about boundary quality. Furthermore, entropy treats mid-stage and boundary frames identically because stages are long and boundaries are brief, mid-stage frames overwhelm the average and move the acquisition signal away from where labelling helps most. Boundary-uncertainty acquisition, by contrast, evaluates directly at the decoded boundary locations and is not diluted by the number of high-confidence mid-stage frames, making it a more useful acquisition signal.

One of the most surprising findings is HistGradientBoosting’s insensitivity to acquisition strategy at 8.59s (random) versus 7.80s (boundary-unc) versus 8.06s (entropy) does not differ significantly at $p \leq 0.05$. This could be explained by HGB’s gradient-boosted ensemble already incorporating implicit diversity through its iterated weak learners, reducing the marginal value of additionally diverse training examples chosen by AL. Furthermore, HGB may have approached the upper bound of what the window-summary feature representation supports on this dataset and no acquisition strategy could then further reduce its boundary error. The neural sequence models on the same data achieving substantially lower MAE (5.9 vs 7.8 s) supports this conclusion that the headroom for AL gains is present in models with sufficient capacity, not in the acquisition function alone.

However, limitations were encountered during this work. The four-stage segmentation reflects one interpretive scheme and all labels were produced by a single annotator, so the reported boundary errors measure agreement with that annotator’s interpretive scheme rather than with any external ground truth. A formal inter-annotator agreement study would establish an empirical ceiling on achievable accuracy for this task. Also, our results are from a single installation with a fixed two-arm setup where the strong performance of template-based methods such as DTW suggests the embrace motions are highly repeatable and the pipeline may transfer less readily to performances with greater structural diversity.

6. Conclusion

We have presented an active-learning pipeline that turns raw robotic-arm performance recordings into a stage-segmented dataset with substantially reduced manual annotation cost. A boundary-uncertainty acquisition function targets ensemble disagreement on decoded boundary locations rather than on per-frame label entropy. A TCN initialised from SSL pretraining and trained with boundary-uncertainty AL outperformed comparative methods. This work is a step toward making data archives usable as resources for downstream AI methods and may transfer to other interpretively-labelled datasets in the arts, health and beyond.

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